

Supply Chain Big Data Project — Dataset Summary

Current scope: four marketplace product datasets — Flipkart, Amazon, Shopee and Lazada — have been collected, cleaned and preprocessed. Alibaba has been removed from the project. Raw files are kept unchanged in `raw/`; cleaned files are in `preprocessed/` and are regenerated by `scripts/preprocess.py`.

Dataset	Raw rows x cols	Clean rows x cols	Clean size	Currency	Cleaned file
Flipkart	20,000 x 15	19,998 x 20	~25 MB	INR	preprocessed/flipkart/flipkart_clean.csv
Amazon	30,000 x 14	30,000 x 20	~71 MB	USD	preprocessed/amazon/amazon_2023_clean.csv
Shopee	30,000 x 52	30,000 x 45	~45 MB	TWD	preprocessed/shopee/shopee_clean.csv
Lazada	30,000 x 11	30,000 x 15	~38 MB	not stated	preprocessed/lazada/lazada_clean.csv

Every cleaned file was re-read and checked: no duplicate product IDs, and one CSV line per record (embedded line breaks were removed from text fields).

Cleaning applied to all four datasets

- Column names standardised to snake_case, with shared names across platforms (product_id, product_name, brand, category_path, category_l1 to category_l3, rating, description) and a **platform** column.
- Text fields: HTML tags removed, whitespace and line breaks collapsed, empty strings turned into missing values.
- Category split into a full path plus the first three levels.
- Placeholder brands such as "No Brand", "nobrand", "OEM" and "Generic" set to missing.
- Duplicate product IDs removed. Missing values are left empty; nothing is imputed.

1. Flipkart — Product Catalog (India)

Source: PromptCloudHQ Flipkart Products, mirrored on Hugging Face from Kaggle.
https://huggingface.co/datasets/jason1966/PromptCloudHQ_flipkart-products

- 2 duplicate product IDs removed (20,000 to 19,998 rows).
- Category tree parsed; 328 rows whose "category" was only a truncated product name were set to missing.
- "No rating available" converted to missing; the two identical rating columns merged into one.
- Product specifications converted from Ruby-hash text to valid JSON.
- Added discount_pct, first image URL and image count; crawl timestamp normalised to UTC.
- Still missing: rating 91%, brand 29%, price 0.4%.

2. Amazon — Amazon Products 2023 (USD)

Source: milistu/AMAZON-Products-2023 on Hugging Face (117,243 products). A random 30,000-row sample (seed 42) was taken across the whole dataset and the embeddings column was dropped.
<https://huggingface.co/datasets/milistu/AMAZON-Products-2023>

- Brand taken from the store field, with "Format: ..." suffixes removed.
- Category list split into path and levels; feature bullets joined into one text field with a feature count.
- Details converted from Python-dict text to valid JSON; zero prices set to missing.
- Added a clean product URL and first-available date.
- Still missing: price 31%, main_category 21%, brand 7%.

The Amazon India listing file that was previously in the folder (10 columns, prices in INR, only 11 of 112 sub-categories) did not match this description and is no longer part of the working set.

3. Shopee — Product Listings (Taiwan)

Source: Shopee Product Listings Dataset from Rebrowser (preview sample), on Hugging Face. All 30,000 rows in our file are Taiwan listings priced in TWD. <https://huggingface.co/datasets/rebrowser/shopee-dataset>

- Dropped 9 unusable columns: 6 redacted to "[PREMIUM]" in the free sample (price, the three before-discount prices, models, productUrl), 2 completely empty (isIndividualSeller, isMart) and the rounded _primaryKey.
- Prices and shipping fee divided by 100,000 to give real TWD (7500000 becomes 75).
- Rating of 0 with no reviews set to missing; shipping fee of -1 (unknown) set to missing.
- Category JSON converted to readable names; the two seller-location columns merged; timestamps normalised.
- Still missing: shipping fee 85%, rating 68%, brand 59%, seller rating 11%.

4. Lazada — Product Catalog (Southeast Asia)

Source: Shopping Companion / Lazada catalog on Hugging Face (about 1.3 million products, ~20 GB). Our working file is a 30,000-row extract. <https://huggingface.co/datasets/yuzhan2205/Shopping-companion>

- Dropped 2 redundant columns: contents (name, attributes and options concatenated) and the duplicate product ID.
- Brand extracted from the attributes JSON; empty option lists set to missing and an option count added.
- The source does not state a currency, so no currency column was added.
- Still missing: brand 68%, options 45%, description 39%, attributes 22%.

Suitability for machine learning

- **Price prediction** (regression) from category, brand, text and seller features — all four datasets.
- **Category classification** from product name and description — all four datasets.
- **Rating / popularity prediction** — best on Shopee (seller metrics, shipping, likes) and Amazon (rating counts).
- **Discount analysis and seller clustering** — Flipkart and Shopee have discount fields; Shopee and Lazada have seller IDs.
- Prices are in different currencies and must be converted or modelled per platform. Shopee text is in Chinese.

Files removed during cleanup

- Hugging Face download caches (lock and metadata files).
- The full 20 GB Lazada catalog download and the Shopee daily snapshot folder; the working raw CSVs extracted from them are kept.
- The amz20 Amazon review dataset (sentiment text, not product data) and the 180 MB Amazon India source file.

Project status: collection, profiling, cleaning, preprocessing and validation are complete for the four-platform scope. Next: feature engineering and model building on the cleaned CSVs.